OFFICIAL ANSWER KEY

NCERT Booster Test Series for NEET-2026_Test-06

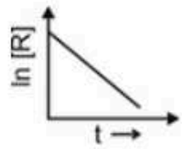
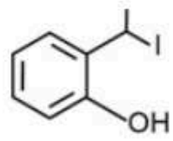
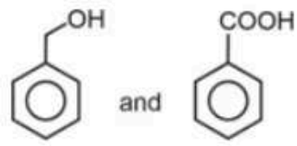
Q.No	Correct Option	Answer
1	B	48 m/s
2	C	$7\mu mg$
3	A	$(\hat{i}+\frac{\hat{j}}{2})$ m/s
4	D	$\frac{m_2}{m_1}$
5	A	20 m/s
6	B	1.0 cm
7	B	960 *K*
8	D	Both Assertion and Reason are false statements
9	D	Compressing the gas through adiabatic process will require more work to be done
10	A	$\frac{d}{\sqrt{3}}$
11	A	236 Hz
12	A	Statement A is true but B is false
13	C	(A)-PS; (B)-QR; (C)-QR; (D)-PS
14	D	$\frac{100}{\sqrt{2}}V$
15	B	Microwaves
16	A	1.4 eV
17	A	13.6 eV
18	A	9×10^{10} J
19	C	2ν
20	C	14.42 mm

21	B	500 V
22	D	$\frac{v}{2}$
23	D	1Ω
24	D	40 cm
25	C	10 V
26	D	Electric current
27	B	3
28	C	$\frac{1}{160\pi}$ rad
29	C	K is negative, it implies if *A* has concave meniscus, then B should have convex meniscus
30	C	$mgx_0 + \frac{1}{2}kx_0^2 + P_0Ax_0 + \frac{1}{2}mv_0^2$
31	D	$\frac{a}{2}\hat{i} + \frac{a}{2\sqrt{3}}\hat{j}$
32	A	The direction of $\vec{r}$
33	B	$\frac{\sigma\pi R^4}{4}$
34	A	Both (A) and (R) are correct and (R) is the correct explanation of A
35	D	$\sqrt{74}$ m/s
36	B	$\frac{3-\sqrt{2}}{3}$
37	D	2
38	D	$\frac{(K_1+K_2)^2}{4K_1K_2}$
39	B	2 watt
40	D	Both (A) and (R) are correct and (R) is the correct explanation of (A)
41	B	$\frac{\sqrt{2}\mu_0 I}{3\pi a}$
42	A	$\frac{2mv}{qB_1}\left[1 - \frac{B_1}{B_2}\right]$
43	C	1 cm
44	B	$\frac{Q}{2\varepsilon_0}$
45	C	$\frac{\alpha^2}{4(1-\alpha)^2}$
46	A	Both Assertion & Reason are true and the reason is the correct explanation of the assertion

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47	A	4 g
48	C	-6.3 kcal mol^{-1}
49	D	Mesitylene
50	D	Th
51	B	40%
52	B	+2
53	B	n = 4, l = –2, m = 2, s = $-\frac{1}{2}$
54	B	
55	D	– 2.1 F
56	A	(a) and (b) only
57	B	A is limiting reactant and 60 molecules of A_3B_2 will be formed
58	D	$MX > MX_3 > MX_2$
59	C	A is true but R is false
60	C	a and c only
61	C	Statement I is correct but statement II is incorrect
62	A	5 and 6
63	C	+2, +6, +4
64	A	(b) and (c) only
65	B	sp^2, 1
66	B	(i) and (iii) only
67	A	
68	D	$C_2H_5OH = Mg_3(PO_4)_2 = KBr = Na_3PO_4$
69	C	0.6 bar
70	A	Both statements I and II are correct

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71	C	
72	B	C_3O_2
73	B	$[CoCl(NH_3)_5]^{2+}$
74	D	Hex-3-yne, cis-Hex-3-ene and Hexan-3, 4-diol
75	A	A and B only
76	C	a(iv), b(i), c(ii), d(iii)
77	B	
78	C	2
79	A	
80	D	(A) is false but (R) is true
81	D	Statement I is incorrect but statement II is correct
82	C	Scarlet red
83	C	H_2O because of hydrogen bonding
84	B	a(iv), b(i), c(ii), d(iii)
85	C	1.57 V
86	C	8
87	D	a(ii), b(iii), c(i), d(iv)
88	C	$[Co(NH_3)_3Cl_3]$
89	C	NO_3^-
90	C	Cell wall of its members are composed of cellulose, galactans, mannans and calcium carbonate.
91	D	(b) and (c) are false
92	C	Seven
93	C	Primary CO_2 acceptor is a four-carbon compound

94	C	One molecule of pyruvic acid, 2 molecules of ATP and one molecule of $NADH + H^+$
95	B	If we plot this type of growth against time, we will get a straight line.
96	B	(a) and (b)
97	C	A gamete has only one chromosome and only one allele.
98	A	Also inhibits abscission of young leaves
99	D	Present in genetic material of *E. coli*
100	B	Biofertilizers are inorganic compound that enrich the nutrient quality of soil.
101	C	Have genetically identical embryo except zygotic one
102	C	A(iii), B(i), C(iv), D(ii)
103	C	5′ GTCCCACATGCAGTG 3′
104	A	(a) and (c)
105	B	A single layer of tall and slender cells with basal nuclei
106	C	Only b and c
107	C	2
108	D	Both 'B' and 'C' are dioecious and free-living.
109	C	Indirect development
110	A	Bell-shaped
111	D	Both (A) and (R) are false
112	C	Only statement (A) is incorrect
113	B	60 sec
114	A	The urine concentrating ability depends upon the levels of ADH in blood plasma
115	C	(A) is true but (R) is false
116	D	1 and 3
117	B	(a), (b), (c), (e) and (f)
118	B	Floral formula represents cohesion and adhesion within the parts of whorls and between the whorls, respectively

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119	C	18.75%
120	A	Both (A) and (R) are true but (R) is not the correct explanation of (A)
121	D	D, B, E, C, A, F
122	C	Only statement I is correct
123	C	Sun flower < *Calotropis* < *Alstonia*
124	B	Statements C and D are true and statement D explains statement C correctly.
125	B	Only (b) and (d)
126	B	E. A, D, B, C
127	B	Prophase
128	B	10,000 kJ
129	B	Cytosolic enzymes are non-functional
130	D	Myasthenia gravis
131	C	Regulation of gastric secretions and respiration
132	D	Both (A) and (R) are false
133	B	Follicular
134	A	Latter one contains steroidal preparation while the former one is non-steroidal.
135	B	*Sall*
136	A	(1)
137	C	The rate of appearance of new forms is not linked with the life cycle or the life span.
138	D	(4)
139	C	Placenta secretes sufficient amounts of progesterone and estrogen
140	B	Three
141	A	Four
142	B	Head is triangular and is formed by the fusion of six embryonic segments.
143	C	Sea hare

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144	D	Limbless body
145	A	Peroxidase
146	B	Moist
147	C	a(ii); b(i); c(iv); d(iii)
148	B	(2)
149	D	Pivot, gliding
150	B	Na^+ influx makes the inside of axonal membrane negative as compared to the outside
151	B	Increasing Ca^{2+} absorption from the digested food
152	A	24
153	D	GIFT
154	C	(b) and (d)
155	A	2 : 1
156	B	Lactate dehydrogenase
157	B	Both (A) and (R) are true but (R) is not the correct explanation of (A)
158	A	In tomato, the enzymes that participate in dark reaction are much sensitive to high temperature than that in maize
159	A	Increase in pH of intermembrane space
160	D	Plant B can be cucumber which is treated with ethylene
161	C	After two generations in ^{14}N medium, only light DNA was observed
162	A	Tissues labelled as D, E and F constitute the stele
163	B	Survive adverse environmental condition and aids transmission
164	C	Birth rate = Death rate
165	B	Classification – Grouping based only on morphological characters
166	D	(4)
167	C	The given food web is composed of more than three food chains and none of them will form spindle shaped pyramid of number
168	B	Four
169	A	Materials to be packaged in the form of vesicles from the endoplasmic reticulum fuse with the concave face of the Golgi apparatus and then

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		move towards the convex face.
170	C	One
171	B	(2)
172	C	Restriction endonuclease cuts each of the two strands of the DNA double helix at specific points in their sugar-phosphate backbones.
173	D	Hormone which is released by pars distalis and regulates growth of mammary glands.
174	A	a(iii); b(iv); c(i); d(ii)
175	D	Only statement (B) is correct
176	B	One
177	C	Pelycosaurs – Descendants of synapsids
178	C	300 mOsmol/L
179	C	Muscle would remain in a rigid and contracted state
180	B	6

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DETAILED SOLUTIONS & EXPLANATIONS

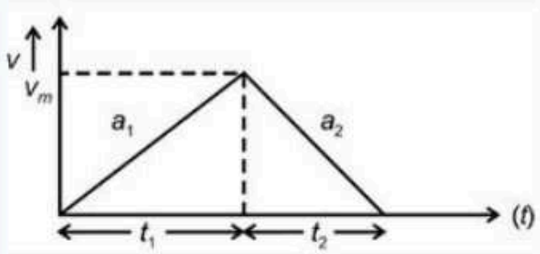
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(B) -

48 m/s

Solution



$a_1 = 2\ \text{m s}^{-2}$, $a_2 = 3\ \text{m s}^{-2}$

$T = 40$ s (total time)

$$a_1 = \frac{v_m}{t_1}, \quad a_2 = \frac{v_m}{t_2}$$

$$T = t_1 + t_2 = \frac{v_m}{a_1} + \frac{v_m}{a_2}$$

$$v_m = \frac{a_1 \times a_2}{a_1 + a_2} T$$

$$\therefore v_m = \frac{2 \times 3}{2 + 3} \times 40 = \frac{6}{5} \times 40 = 48\ \text{m/s}$$

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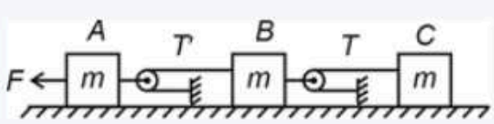
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(C) -

$7\mu mg$

Solution



$T \geq \mu mg$

$T' \geq \mu mg + 2T$

$F \geq 2T' + \mu mg$

Hence $F_{min} = 7\mu mg$

3

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(A) - $\left(\hat{i}+\frac{\hat{j}}{2}\right)\ \mathrm{m/s}$

Solution

Applying conservation of linear momentum,

$2\left(2\hat{i}+\hat{j}\right)+4\left(0\right)=2\left(0\right)+4\vec{v}$

$\vec{v}=\left(\hat{i}+\frac{\hat{j}}{2}\right)\ \mathrm{m/s}$

4

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(D) -

$\frac{m_2}{m_1}$

Solution

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$v=\sqrt{\frac{G}{\left(m_1+m_2\right)r}}\times m_2$

$K_1=\frac{1}{2}m_1v_1^2=\frac{1}{2}m_1\frac{G}{\left(m_1+m_2\right)r}m_2^2$

$K_2=\frac{1}{2}m_2\frac{G}{\left(m_1+m_2\right)r}m_1^2$

$\frac{K_1}{K_2}=\frac{m_1m_2^2}{m_2m_1^2}=\frac{m_2}{m_1}$

5

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(A) -

20 m/s

Solution

Hint :

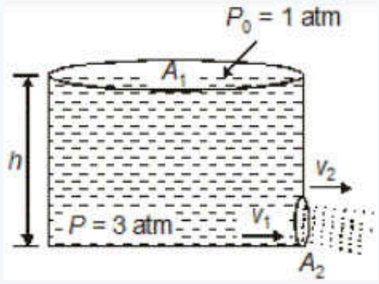
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Velocity of efflux v = $\sqrt{\frac{2gh}{1\u2009-\u2009\frac{A_2^2}{A_1^2}}}$

\r\n\r\n

Solution :

\r\n\r\n



\r\n\r\n

$h = \frac{P\u2009-\u2009P_0}{\rho g}\u2009 = \u2009\frac{2\u2009\times\u2009\u200910^5}{10^4} = 20\u2009cm$.

\r\n\r\n

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Now from Bernoulli\'s theorem.

\r\n\r\n

$P_0 + \rho gh + \frac{1}{2}\rho v_1^2\u2009=\u2009P_0\u2009+\u2009\frac{1}{2}\rho v_2^2$

\r\n\r\n

from equation of continuity.

\r\n\r\n

$A_1v_1 = A_2v_2$

\r\n\r\n

$v_1 = \frac{A_2}{A_1}\cdot v_2\u2009 = \u2009\frac{\sqrt{3}}{2}v_2$

\r\n\r\n

$v_2 = \sqrt{\frac{2gh}{1\u2009-\u2009(\frac{A_2}{A_1})^2}}\u2009 = \u2009\sqrt{\frac{2\u2009\times\u2009\u200910\u2009\times\u2009\u200920}{1\u2009-\u2009(\frac{3}{4})}} = \sqrt{\frac{400}{\frac{1}{4}}} = 2 \times 20$

\r\n\r\n

= 40 m/s

\r\n\r\n

$v_2\u2009=\u200940\u2009m/s$

\r\n

6

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(B) -

1.0 cm

Solution

$$\Delta L = L\alpha\Delta T$$
$$\r\n\Delta L = \left(10\ \text{m}\right)\left(10 \times 10^{-6}/^{\circ}\text{C}\right)\left(100^{\circ}\text{C} - 0^{\circ}\text{C}\right)$$
$$\r\n\Delta L = 10^{-2}\ \text{m}$$
$$\r\n\Delta L = 1\ \text{cm}$$
$$\r\n$$

7

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(B) -

960 *K*

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Solution

$$\eta_1 = \eta_2$$
$$1 - \frac{500}{300} = 1 - \frac{600}{x_{120}}$$
$$= x = \frac{600^{\circ} \times 800^{\circ}}{500^{\circ}} = 960\ \text{K}$$
$$\r\n$$

8

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(D) -

Both Assertion and Reason are false statements

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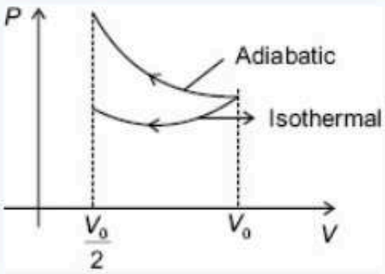
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(D) -

Compressing the gas through adiabatic process will require more work to be done

Solution



Area under $P - V$ curve of adiabatic process is more therefore work-done requires more.

10

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(A) -

$\frac{d}{\sqrt{3}}$

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Solution

Hint :

Use conservation of energy

Solution :

$\frac{1}{2}kd^2 = \frac{1}{2}m\left(A\omega\right)^2 + \frac{1}{2}2m\left(A\omega\right)^2$

$\frac{1}{2}kd^2 = \frac{1}{2}kA^2 + kA^2$

$3A^2 = d^2$

$A = \frac{d}{\sqrt{3}}$

11

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(A) -

236 Hz

Solution

$n = \frac{1}{2l}\sqrt{\frac{T}{m}}$

$n \propto \frac{1}{l}$

$n - n_1 = 4$

$n_1 = n - 4$

$n_2 = n + 4$

$\frac{n_1}{n_2} = \frac{l_2}{l_1} = \frac{n-4}{n+4}$

$\frac{58}{60} = \frac{n-4}{n+4}$

n = 236 Hz

12

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(A) -

Statement A is true but B is false

Solution

For air medium

v_0 = 330 m/s

$v_t = v_0 + \frac{v_0 \times t}{546}$

$v_t = v_0 + \frac{330}{546}t$

$= v_0 + 0.61t$

13

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(C) -

(A)-PS; (B)-QR; (C)-QR; (D)-PS

Solution

When current *I* in wire is increased, the inward flux in loop increases, hence an anticlockwise current will be induced and wire repels the loop. When current *I* decreases, then loop will be attracted by the wire and a clockwise current will be induced in loop.
Similarly, when loop is moved towards wire, flux increases then by Lenz's law anticlockwise current will be induced and when loop is moved away then clockwise current will be induced in the loop.

14

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(D) -

$\frac{100}{\sqrt{2}}\mathrm{V}$

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15

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(B) -

Microwaves

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16

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(A) -

1.4 eV

Solution

$K_{max} = E - W_0 = 3.4 - 2 = 1.4$ eV

17

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(A) -

13.6 eV

Solution

Kinetic energy (K.E.) = –(Total energy)

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K.E. = –(–13.6) = 13.6 eV

18

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(A) -

9×10^{10} J

Solution

Einstein's mass-energy equivalence relation, $E = mc^2 = 10^{-6} \times (3 \times 10^8)^2 = 9 \times 10^{10}$ J

19

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(C) -

2 ν

Solution

$\nu_{output} = 2\nu_{input}$ for full wave rectifier

20

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(C) -

14.42 mm

Solution

Reading = MSR + CSR

= 14 + $\frac{21}{50}$

= 14.42 mm

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21

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(B) -

500 V

Solution

$$\int_{V}^{} dv = -\int E_x dx$$

$$\int_{0}^{V} dv = -10\int_{10}^{0} x\,dx = 10 \times \frac{10^2}{2} = 500\ V$$

22

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(D) -

$\frac{v}{2}$

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23

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(D) -

1Ω

Solution

Answer (4)

\r\n

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24

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(D) -

40 cm

Solution

The equivalent focal length of the mirror:

$\frac{1}{-f_m} = \frac{2}{f_L} - \frac{1}{f_{mirror}}$

$\frac{1}{-f_m} = \frac{2}{f_L} - \frac{1}{\infty}$

$f_m = \frac{-f_L}{2}$ ...(i)

Since the image is formed on the object itself, it means the object is placed on the centre of curvature of the equivalent concave mirror,

Hence $-2f_m = 40$

$-2 \times \frac{-f_L}{2} = 40$

$\therefore f_L = 40$ cm

25

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(C) -

10 V

Solution

$i = 2 + t$

$E = Ri + L\frac{di}{dt}$

$E = 2(2 + t) + 2\frac{d}{dt}(2 + t)$

$= 4 + 2t + 2$

$E = 6 + 2t = (6 + 2 \times 2)$

$= 10$ V

26

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(D) -

Electric current

Solution

As we know displacement current Through an area is given by $I_d = \epsilon_0 \left(\frac{d\Phi_E}{dt} \right) \rightarrow$ Electric current.

27

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(B) -

3

Solution

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As we know,

$\text{Tan}\theta = \frac{4H}{R}$ [using basic results of H and R]

$\Rightarrow \text{Tan}\theta = \frac{4H}{4\sqrt{3}H} = \frac{1}{\sqrt{3}}$

$\Rightarrow \theta = 30^\circ$

Now $R = \frac{u^2 \sin 2\theta}{g} = \frac{u^2 \times 2\sin\theta\cos\theta}{g}$

$= \frac{u^2 \times 2\sin 30 \cos 30^\circ}{g} = \frac{u^2 \times 2 \times \frac{1}{2} \times \frac{\sqrt{3}}{2}}{g}$

$= \frac{\sqrt{3}u^2}{2g} \Rightarrow n = 3$

28

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(C) -

$\frac{1}{160\pi}$ rad

Solution

Shear modulus = $\frac{\text{Shear stress}}{\text{Shear strain}}$

$G = \frac{F_t / A}{\triangle l / L} = \frac{F_t}{A(\theta)}$

$\theta = \frac{F_t}{GA} = \frac{10^5}{10^{10} \times \pi \times (4 \times 10^{-2})^2}$

$= \frac{10^5}{10^{10} \times \pi \times 16 \times 10^{-4}} = \frac{1}{160\pi}$

29

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(C) -

K is negative, it implies if *A* has concave meniscus, then B should have convex meniscus

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Solution

Answer (3)

\r\n

30

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(C) -

$mgx_0 + \frac{1}{2}kx_0^2 + P_0Ax_0 + \frac{1}{2}mv_0^2$

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31

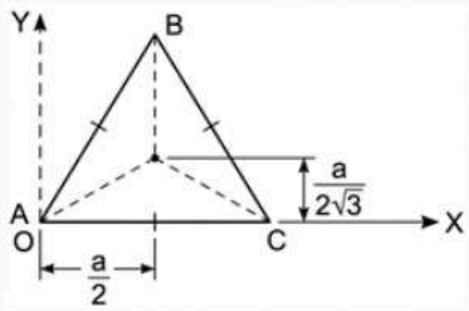
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(D) -

$\frac{a}{2}\hat{i} + \frac{a}{2\sqrt{3}}\hat{j}$

Solution

Here Geometrical centre i.e. centroid of triangle is centre of mass.



32

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(A) -

The direction of $\vec{r}$

Solution

Answer (1)

\r\n

33

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(B) -

$\frac{\sigma\pi R^4}{4}$

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34

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(A) -

Both (A) and (R) are correct and (R) is the correct explanation of A

Solution

Areal velocity of a planet around Sun $\frac{dA}{dt} = \frac{L}{2m}$ here L → angular momentum is constant because gravitational force is central force. $\frac{dA}{dt} \rightarrow$ Areal velocity is constant

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35

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(D) -

$\sqrt{74}$ m/s

Solution

Using work-energy Theorem:

$$\Delta K = W_{all}$$

$$\Rightarrow \tfrac{1}{2}mv^2 - \tfrac{1}{2}mu^2 = -\int_1^{1.9} 10x\,dx$$

$$\Rightarrow v^2 - u^2 = -2 \times 10\left[\tfrac{x^2}{2}\right]_1^{1.9}$$

$$v^2 = u^2 - 10\left[(1.9)^2 - 1^2\right]$$

$$v^2 = 100 - 10\left[3.61 - 1\right]$$

$$= 100 - 10 \times 2.61 = 100 - 26.1$$

$$= 100 - 26 \approx 74 \quad v = \sqrt{74}$$

36

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(B) -

$\frac{3-\sqrt{2}}{3}$

Solution

Answer (2)

\r\n

37

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(D) -

2

Solution

Here $Q_1 = \sigma \times 4\pi R^2$

$Q_2 = \sigma \times 4\pi \left(2R\right)^2$

Total charge of system

$Q = 20\sigma\pi R^2$ (1)

Now when they are in contact.

Their potentials will be same

$\Rightarrow \frac{Q_1^{\prime}}{R_1} = \frac{Q_2^{\prime}}{R_2} \Rightarrow \frac{Q_1^{\prime}}{Q_2^{\prime}} = \frac{R_1}{R_2} = \frac{1}{2}$ (II)

From (I) and (II)

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$Q_1^{\prime} = \frac{20\sigma\pi R^2}{3}$ and $Q_2^{\prime} = \frac{40\sigma\pi R^2}{3}$

38

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(D) -

$\frac{\left(K_1 + K_2\right)^2}{4K_1K_2}$

Solution

Answer (4)

\r\n

39

SAD_LYFFFF

(B) -

2 watt

Solution

Answer (2)

\r\n

40

SAD_LYFFFF

(D) -

Both (A) and (R) are correct and (R) is the correct explanation of (A)

SAD_LYFFFF

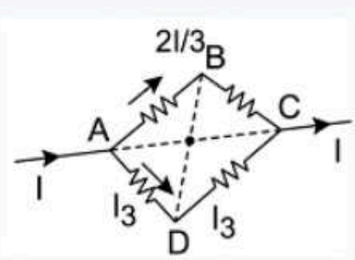
41

SAD_LYFFFF

(B) -

$\frac{\sqrt{2}\mu_0 I}{3\pi a}$

Solution



$$(B_0)_{ABC} = \frac{\mu_0 I \times 2 \times 2\sin 45 \times 2 \quad Cx}{4\pi (r/2) \quad 3} \rightarrow$$

$$(B_0)_{ADC} = \frac{\mu_0 \times I}{4\pi \quad 3(r/2)} \times 2\sin 45 \times 2 (\cdot)$$

42

SAD_LYFFFF

(A) -

$\frac{2mv}{qB_1}\left[1-\frac{B_1}{B_2}\right]$

Solution

Answer (1)

\r\n

43

SAD_LYFFFF

(C) -

1 cm

Solution

SAD_LYFFFF

Answer (3)

\r\n

44

SAD_LYFFFF

(B) -

$\frac{Q}{2\varepsilon_0}$

Solution

$\phi=\frac{Q}{\varepsilon_0}-\frac{Q}{4\pi\varepsilon_0}2\pi\left[1-\cos 60\right]X2=\frac{Q}{\varepsilon_0}-\frac{Q}{2\varepsilon_0}\left[1-\frac{1}{2}\right]X2=\frac{Q}{2\varepsilon_0}$

45

SAD_LYFFFF

(C) -

$\frac{\alpha^2}{4(1-\alpha)^2}$

Solution

Hint :

$K_c = \frac{[H_2]\ [I_2]}{[HI]^2}$

Solution :

	$2HI(g)$	$\rightleftharpoons H_2(g)$	$+ I_2(g)$
Initially	2	0	0
At equilibrium	$2-2\alpha$	α	α

$K_c = \frac{(\alpha)^2}{(2-2\alpha)^2}$

$K_c = \frac{\alpha^2}{4(1-\alpha)^2}$

46

SAD_LYFFFF

SAD_LYFFFF

(A) -

Both Assertion & Reason are true and the reason is the correct explanation of the assertion

Solution

Due to small size of F atom electron gain enthalpy of F is less negative than Cl.

47

SAD_LYFFFF

(A) -

4 g

Solution

Number of moles of HCl = $\frac{320 \times 0.25}{1000} = 0.08$

\r\n\r\n

Moles of $CaCO_3$ required to react with HCl = 0.04

\r\n\r\n

$CaCO_3 + 2HCl \rightarrow CaCl_2 + H_2O + CO_2$

\r\n\r\n

Mass of $CaCO_3$ = 0.04 × 100 = 4g

\r\n

48

SAD_LYFFFF

SAD_LYFFFF

(C) - –6.3 kcal mol^{-1}

Solution

Hint :

$\Delta G = \Delta H - T\Delta S$.

Solution :

$\Delta G = \Delta H - T\Delta S$

$= 5.7 - 300 \times 40 \times 10^{-3}$

$= 5.7 - 12 = -6.3$ kcal mol^{-1}

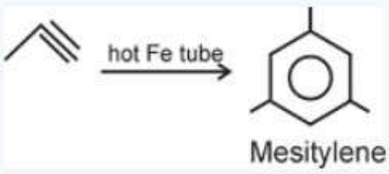
49

SAD_LYFFFF

(D) -

Mesitylene

Solution



50

SAD_LYFFFF

(D) -

Th

SAD_LYFFFF

Solution

Hint :

Actinoids are *f*-block elements in which last electron enters into 5*f* subshell

Solution :

Cu → *d*-block

Te → *p*-block

Ce → Lanthanoid

Th → Actinoid

51

SAD_LYFFFF

(B) -

40%

Solution

Mass of AgBr = 0.47 g

Mole of AgBr = Mole of Br = $\frac{0.47}{188}$

= 2.5×10^{-3}

Mass of Br = $80 \times 2.5 \times 10^{-3}$

= 0.2 g

$\% \text{ Br} = \frac{0.2}{0.5} \times 100$

= 40%

52

SAD_LYFFFF

(B) -

SAD_LYFFFF

+2

Solution

Due to inert pair effect, the most stable oxidation state of Pb is +2.

53

SAD_LYFFFF

(B) -

n = 4, l = –2, m = 2, s = $-\frac{1}{2}$

Solution

Hint :

Negative value of l is not possible.

Solution :

For n = 4, the possible value of l are 0, 1, 2 and 3 only.

54

SAD_LYFFFF

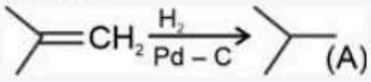
(B) - 

SAD_LYFFFF

Solution

Hint :

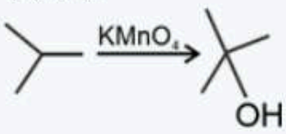
\r\n\r\n



\r\n\r\n

Solution :

\r\n\r\n



\r\n

55

SAD_LYFFFF

(D) -

– 2.1 F

Solution

Hint :

$\Delta_r G^\circ = -nFE^\circ_{cell}$

Solution :

$\Delta_r G^\circ = -2F\ (\ 1\ .05\ )$

$= -2.1\ F$

56

SAD_LYFFFF

(A) -

(a) and (b) only

SAD_LYFFFF

Solution

The transition metal ions which do not contain *d*-electrons are colourless.

57

SAD_LYFFFF

(B) -

A is limiting reactant and 60 molecules of A_3B_2 will be formed

Solution

A is limiting reactant and 60 molecules of A_3B_2 will be formed.

58

SAD_LYFFFF

(D) -

$MX > MX_3 > MX_2$

Solution

$MX\ S_1 = \sqrt{4 \times 10^{-8}} = 2 \times 10^{-4}\ \ mol/l$

$MX_2\ K_{sp} = 4s^3$

$s_2 = \frac{3.2 \times 10^{-14\frac{1}{3}}}{4} = 2 \times 10^{-6}\ mol/l$

$MX_3\ \ K_{sp} = 27s^4$

$s_3 = \frac{K_{sp}{}^{\frac{1}{4}}}{27} = \frac{2.7 \times 10^{-15\frac{1}{4}}}{27}$

$s_3 = 3.16 \times 10^{-4}\ mol/l$

59

SAD_LYFFFF

(C) -

A is true but R is false

SAD_LYFFFF

Solution

Energies of the orbitals in the same subshell decreases with increase in the atomic number.

60

SAD_LYFFFF

(C) -

a and c only

Solution

N_2O is not an odd electron molecule.

61

SAD_LYFFFF

(C) -

Statement I is correct but statement II is incorrect

Solution

Hint :

SiF_6^{2-} is known.

Solution :

$SiCl_6^{2-}$ is not known because six large chloride ions cannot be accommodated around Si^{4+} due to the crowding.

62

SAD_LYFFFF

(A) -

5 and 6

SAD_LYFFFF

Solution

Hint :

Electronic configuration of O_2:

$(\sigma 1s)^2\ (\sigma^* 1s)^2\ (\sigma 2s)^2\ (\sigma^* 2s)^2\ (\sigma 2p_z)^2\ \ \pi 2p_x^2 = \pi 2p_y^2 \pi^* 2p_x^1 = \pi^* 2p_y^1$

Solution :

Electronic configuration of

O_2^+ : $(\sigma 1s)^2\ (\sigma^* 1s)^2\ (\sigma 2s)^2\ (\sigma^* 2s)^2\ (\sigma 2p_z)^2\ \ \pi 2p_x^2 = \pi 2p_y^2 \pi^* 2p_x^1$

B_2 : $(\sigma 1s)^2\ (\sigma^* 1s)^2\ (\sigma 2s)^2\ (\sigma^* 2s)^2\ \ \pi 2p_x^1 = \pi 2p_y^1$

63

SAD_LYFFFF

(C) - +2, +6, +4

Solution

Hint :

For neutral molecules, net charge = 0.

Solution :

$Na_2S_2O_3 \Rightarrow 2 \times 1 + 2 \times a + 3x(-2) = 0 \Rightarrow a = 2$

$SF_6 \Rightarrow a + 6x(-1) \Rightarrow a = 6$

$SO_2 \Rightarrow a + 2x(-2) \Rightarrow a = 4$

64

SAD_LYFFFF

(A) -

(b) and (c) only

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Solution

Gallium is also known as Eka-aluminium.

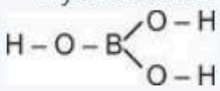
65

SAD_LYFFFF

(B) - sp^2, 1

Solution

→ Hybridisation of B is sp^2,



→ H_3BO_3 does not lose $3H^+$ ion but accepts one lone pair from H_2O molecule (or hydroxide ion).

∴ It is a monobasic acid.

66

SAD_LYFFFF

(B) -

(i) and (iii) only

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Solution

- Resonance effect work along π electrons.
- Inductive effect is a permanent effect.

67

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(A) - 

Solution

Hint :

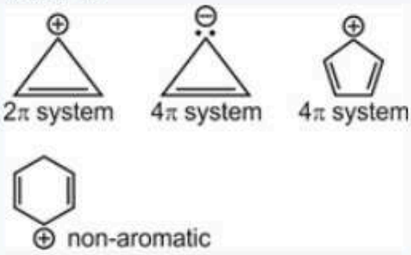
\r\n\r\n

Cyclic planar structures having (4n + 2)π delocalized electrons are aromatic.

\r\n\r\n

Solution :

\r\n\r\n



\r\n

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68

SAD_LYFFFF

(D) -

$C_2H_5OH = Mg_3(PO_4)_2 = KBr = Na_3PO_4$

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69

SAD_LYFFFF

(C) -
0.6 bar

SAD_LYFFFF

70

SAD_LYFFFF

(A) -
Both statements I and II are correct

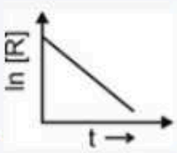
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SAD_LYFFFF

71

SAD_LYFFFF

(C) -



SAD_LYFFFF

72

SAD_LYFFFF

(B) -

C_3O_2

SAD_LYFFFF

73

SAD_LYFFFF

(B) -

$[CoCl(NH_3)_5]^{2+}$

SAD_LYFFFF

SAD_LYFFFF

74

SAD_LYFFFF

(D) -

Hex-3-yne, cis-Hex-3-ene and Hexan-3, 4-diol

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75

SAD_LYFFFF

(A) -
A and B only

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76

SAD_LYFFFF

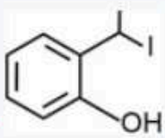
(C) -
a(iv), b(i), c(ii), d(iii)

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77

SAD_LYFFFF

(B) - 

SAD_LYFFFF

78

SAD_LYFFFF

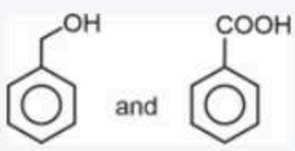
(C) -

2

SAD_LYFFFF

79

SAD_LYFFFF

(A) - 

SAD_LYFFFF

SAD_LYFFFF

80

SAD_LYFFFF

(D) -

(A) is false but (R) is true

SAD_LYFFFF

81

SAD_LYFFFF

(D) -

Statement I is incorrect but statement II is correct

SAD_LYFFFF

82

SAD_LYFFFF

(C) -

Scarlet red

SAD_LYFFFF

SAD_LYFFFF

83

SAD_LYFFFF

(C) -

H_2O because of hydrogen bonding

SAD_LYFFFF

84

SAD_LYFFFF

(B) -
a(iv), b(i), c(ii), d(iii)

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85

SAD_LYFFFF

(C) -
1.57 V

SAD_LYFFFF

SAD_LYFFFF

86

SAD_LYFFFF

(C) -
8

SAD_LYFFFF

87

SAD_LYFFFF

(D) -

a(ii), b(iii), c(i), d(iv)

SAD_LYFFFF

88

SAD_LYFFFF

(C) -

$[Co(NH_3)_3Cl_3]$

SAD_LYFFFF

SAD_LYFFFF

89

SAD_LYFFFF

(C) -

NO_3^-

Solution

$$NO_3^- + H_2SO_4 \rightarrow HSO_4^- + HNO_3$$

$$4HNO_3 \rightarrow 4NO_2 + O_2 + 2H_2O$$

$$2NO_3^- + 4H_2SO_4 + 3Cu \rightarrow 3CuSO_4 + SO_4^{2-} + 4H_2O + 2NO$$

$$2NO + O_2 \rightarrow \underset{\text{(Brown fumes)}}{2NO_2}$$

90

SAD_LYFFFF

(C) -

Cell wall of its members are composed of cellulose, galactans, mannans and calcium carbonate.

Solution

NCERT Reference: XI, Page No. 11, 15, 16

91

SAD_LYFFFF

(D) -

(b) and (c) are false

Solution

SAD_LYFFFF

NCERT Reference: XI, Page No. 74

92

SAD_LYFFFF

(C) -

Seven

Solution

NCERT Reference: XI, Page No. 28, 29, 31, 32, 33

93

SAD_LYFFFF

(C) -

Primary CO_2 acceptor is a four-carbon compound

Solution

NCERT Reference: XI, Page No. 145, 146, 147

94

SAD_LYFFFF

(C) -

One molecule of pyruvic acid, 2 molecules of ATP and one molecule of NADH + H^+

Solution

NCERT Reference: XI, Page No. 156, 157

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95

SAD_LYFFFF

(B) -

If we plot this type of growth against time, we will get a straight line.

Solution

NCERT Reference: XI, Page No. 169, 170

96

SAD_LYFFFF

(B) -

(a) and (b)

Solution

NCERT Reference: XI, Page No. 76, 77

97

SAD_LYFFFF

(C) -

A gamete has only one chromosome and only one allele.

Solution

NCERT Reference: XII, Page No. 65 SAD_LYFFFF

98

SAD_LYFFFF

(A) -

Also inhibits abscission of young leaves

Solution

NCERT Reference: XI, Page No. 175, 176

99

SAD_LYFFFF

(D) -

Present in genetic material of *E. coli*

Solution

NCERT Reference: XII, Page No. 80, 81

100

SAD_LYFFFF

(B) -

Biofertilizers are inorganic compound that enrich the nutrient quality of soil.

Solution

NCERT Reference: XII, Page No. 154, 155, 157, 158

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101

SAD_LYFFFF

(C) -

Have genetically identical embryo except zygotic one

Solution

NCERT Reference: XII, Page No. 23, 22

102

SAD_LYFFFF

(C) -

A(iii), B(i), C(iv), D(ii)

Solution

NCERT Reference: XI, Page No. 24, 29, 33, 34

103

SAD_LYFFFF

(C) -

5′ GTCCCACATGCAGTG 3′

Solution

NCERT Reference: XI, Page No. 96

SAD_LYFFFF

104

SAD_LYFFFF

(A) -

(a) and (c)

Solution

NCERT Reference: XII, Page No. 7

105

SAD_LYFFFF

(B) -

A single layer of tall and slender cells with basal nuclei

Solution

NCERT Reference: XI, Page No. 101 (old)

106

SAD_LYFFFF

(C) -

Only b and c

Solution

NCERT Reference: XI, Page No. 114 (old)

SAD_LYFFFF

107

SAD_LYFFFF

(C) -

2

Solution

NCERT Reference: XI, Page No. 83

108

SAD_LYFFFF

(D) -
Both 'B' and 'C' are dioecious and free-living.

Solution

NCERT Reference: XI, Page No. 43, 44

109

SAD_LYFFFF

(C) -
Indirect development

Solution

NCERT Reference: XI, Page No. 46SAD_LYFFFF

110

SAD_LYFFFF

(A) -
Bell-shaped

Solution

NCERT Reference: XI, Page No. 116

111

SAD_LYFFFF

(D) -

Both (A) and (R) are false

Solution

NCERT Reference: XI, Page No. 112

112

SAD_LYFFFF

(C) -

Only statement (A) is incorrect

Solution

NCERT Reference: XI, Page No. 188

SAD_LYFFFF

113

SAD_LYFFFF

(B) -

60 sec

Solution

NCERT Reference: XI, Page No. 200

114

SAD_LYFFFF

(A) -

The urine concentrating ability depends upon the levels of ADH in blood plasma

Solution

NCERT Reference: XI, Page No. 212

115

SAD_LYFFFF

(C) -

(A) is true but (R) is false

Solution

NCERT Reference: XII, Page No. 67, 68

SAD_LYFFFF

116

SAD_LYFFFF

(D) -

1 and 3

Solution

NCERT Reference: XII, Page No. 10, 11

117

SAD_LYFFFF

(B) -

(a), (b), (c), (e) and (f)

Solution

NCERT Reference: XII, Page No. 29, 30, 32

118

SAD_LYFFFF

(B) -

Floral formula represents cohesion and adhesion within the parts of whorls and between the whorls, respectively

Solution

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NCERT Reference: XII, Page No. 68

119

SAD_LYFFFF

(C) -

18.75%

Solution

NCERT Reference: XII, Page No. 57

120

SAD_LYFFFF

(A) -

Both (A) and (R) are true but (R) is not the correct explanation of (A)

Solution

NCERT Reference: XII, Page No. 76

121

SAD_LYFFFF

(D) -

D, B, E, C, A, F

Solution

NCERT Reference: XII, Page No. 90, 91, 92, 98, 99

SAD_LYFFFF

122

SAD_LYFFFF

(C) -

Only statement I is correct

Solution

NCERT Reference: XII, Page No. 188, 198

123

SAD_LYFFFF

(C) -

Sun flower < *Calotropis* < *Alstonia*

Solution

NCERT Reference: XII, Page No. 61

124

SAD_LYFFFF

(B) -

Statements C and D are true and statement D explains statement C correctly.

Solution

NCERT Reference: XII, Page No. 218, 220, 221

SAD_LYFFFF

125

SAD_LYFFFF

(B) -

Only (b) and (d)

Solution

NCERT Reference: XI, Page No. 93

126

SAD_LYFFFF

(B) -

E. A, D, B, C

Solution

NCERT Reference: XI, Page No. 33

127

SAD_LYFFFF

(B) -

Prophase

Solution

NCERT Reference: XI, Page No. 121

SAD_LYFFFF

128

SAD_LYFFFF

(B) -

10,000 kJ

Solution

NCERT Reference: XII, Page No. 213

129

SAD_LYFFFF

(B) -

Cytosolic enzymes are non-functional

Solution

NCERT Reference: XI, Page No. 160

130

SAD_LYFFFF

(D) -

Myasthenia gravis

Solution

NCERT Reference: XI, Page No. 227

SAD_LYFFFF

131

SAD_LYFFFF

(C) -

Regulation of gastric secretions and respiration

Solution

NCERT Reference: XI, Page No. 236

132

SAD_LYFFFF

(D) -

Both (A) and (R) are false

Solution

NCERT Reference: XI, Page No. 245

133

SAD_LYFFFF

(B) -

Follicular

Solution

NCERT Reference: XII, Page No. 34

SAD_LYFFFF

134

SAD_LYFFFF

(A) -

Latter one contains steroidal preparation while the former one is non-steroidal.

Solution

NCERT Reference: XII, Page No. 45

135

SAD_LYFFFF

(B) -

*Sal*I

Solution

NCERT Reference: XII, Page No. 169

136

SAD_LYFFFF

(A) -

(1)

Solution

NCERT Reference: XII, Page No. 178

SAD_LYFFFF

137

SAD_LYFFFF

(C) -

The rate of appearance of new forms is not linked with the life cycle or the life span.

Solution

NCERT Reference: XII, Page No. 121

138

SAD_LYFFFF

(D) -
(4)

Solution

NCERT Reference: XII, Page No. 143

139

SAD_LYFFFF

(C) -
Placenta secretes sufficient amounts of progesterone and estrogen

Solution

NCERT Reference: XII, Page No. 37

SAD_LYFFFF

140

SAD_LYFFFF

(B) -
Three

Solution

NCERT Reference: XII, Page No. 166

141

SAD_LYFFFF

(A) -

Four

Solution

NCERT Reference: XI, Page No. 103 (old)

142

SAD_LYFFFF

(B) -

Head is triangular and is formed by the fusion of six embryonic segments.

Solution

NCERT Reference: XI, Page No. 112 (old)

SAD_LYFFFF

143

SAD_LYFFFF

(C) -

Sea hare

Solution

NCERT Reference: XI, Page No. 45

144

SAD_LYFFFF

(D) -
Limbless body

Solution

NCERT Reference: XI, Page No. 48, 49

145

SAD_LYFFFF

(A) -
Peroxidase

Solution

NCERT Reference: XI, Page No. 118

SAD_LYFFFF

146

SAD_LYFFFF

(B) -
Moist

Solution

NCERT Reference: XI, Page No. 183

147

SAD_LYFFFF

(C) -

a(ii); b(i); c(iv); d(iii)

Solution

NCERT Reference: XI, Page No. 200

148

SAD_LYFFFF

(B) -

(2)

Solution

NCERT Reference: XI, Page No. 213

SAD_LYFFFF

149

SAD_LYFFFF

(D) -

Pivot, gliding

Solution

Pivot joint is present between atlas and axis.

150

SAD_LYFFFF

(B) -

Na^+ influx makes the inside of axonal membrane negative as compared to the outside

Solution

NCERT Reference: XI, Page No. 233, 234

151

SAD_LYFFFF

(B) -

Increasing Ca^{2+} absorption from the digested food

Solution

NCERT Reference: XI, Page No. 242

SAD_LYFFFF

152

SAD_LYFFFF

(A) -

24

Solution

NCERT Reference: XII, Page No. 38

153

SAD_LYFFFF

(D) -
GIFT

Solution

NCERT Reference: XII, Page No. 48

154

SAD_LYFFFF

(C) -
(b) and (d)

Solution

NCERT Reference: XII, Page No. 117, 118

SAD_LYFFFF

155

SAD_LYFFFF

(A) -
2 : 1

Solution

NCERT Reference: XII, Page No. 220

156

SAD_LYFFFF

(B) -

Lactate dehydrogenase

Solution

NCERT Reference: XI, Page No. 157

157

SAD_LYFFFF

(B) -

Both (A) and (R) are true but (R) is not the correct explanation of (A)

Solution

NCERT Reference: XI, Page No. 157

SAD_LYFFFF

158

SAD_LYFFFF

(A) -

In tomato, the enzymes that participate in dark reaction are much sensitive to high temperature than that in maize

Solution

NCERT Reference: XI, Page No. 150

\r\n

159

SAD_LYFFFF

(A) -

Increase in pH of intermembrane space

Solution

NCERT Reference: XI, Page No. 160

160

SAD_LYFFFF

(D) -

Plant B can be cucumber which is treated with ethylene

Solution

NCERT Reference: XII, Page No. 177

SAD_LYFFFF

161

SAD_LYFFFF

(C) -

After two generations in ^{14}N medium, only light DNA was observed

Solution

NCERT Reference: XII, Page No. 89

162

SAD_LYFFFF

(A) -

Tissues labelled as D, E and F constitute the stele

Solution

NCERT Reference: XI, Page No. 74

163

SAD_LYFFFF

(B) -

Survive adverse environmental condition and aids transmission

Solution

NCERT Reference: XI, Page No. 15SAD_LYFFFF

164

SAD_LYFFFF

(C) -

Birth rate = Death rate

Solution

NCERT Reference: XII, Page No. 193, 194, 195

165

SAD_LYFFFF

(B) -

Classification – Grouping based only on morphological characters

Solution

NCERT Reference: XI, Page No. 5, 6

166

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(D) -

(4)

Solution

NCERT Reference: XII, Page No. 72, 73, 74, 75

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167

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(C) -

The given food web is composed of more than three food chains and none of them will form spindle shaped pyramid of number

Solution

NCERT Reference: XII, Page No. 211, 212, 213

168

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(B) -
Four

Solution

NCERT Reference: XI, Page No. 121, 122, 123, 124

169

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(A) -
Materials to be packaged in the form of vesicles from the endoplasmic reticulum fuse with the concave face of the Golgi apparatus and then move towards the convex face.

Solution

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NCERT Reference: XI, Page No. 95, 96 97

170

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(C) -
One

Solution

NCERT Reference: XII, Page No. 131

171

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(B) -
(2)

Solution

NCERT Reference: XII, Page No. 182

172

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(C) -
Restriction endonuclease cuts each of the two strands of the DNA double helix at specific points in their sugar-phosphate backbones.

Solution

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NCERT Reference: XII, Page No. 166

173

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(D) -
Hormone which is released by pars distalis and regulates growth of mammary glands.

Solution

NCERT Reference: XII, Page No. 38

174

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(A) -

a(iii); b(iv); c(i); d(ii)

Solution

NCERT Reference: XI, Page No. 49, 50, 51

175

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(D) -

Only statement (B) is correct

Solution

NCERT Reference: XII, Page No. 174, 175

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176

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(B) -

One

Solution

NCERT Reference: XI, Page No. 112

177

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(C) -

Pelycosaurs – Descendants of synapsids

Solution

NCERT Reference: XII, Page No. 123

178

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(C) -

300 mOsmol/L

Solution

NCERT Reference: XI, Page No. 211

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179

SAD_LYFFFF

(C) -

Muscle would remain in a rigid and contracted state

Solution

NCERT Reference: XI, Page No. 222

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(B) -

6

Solution

$$y = \frac{2\lambda D}{d}$$

$$\frac{\partial y}{\partial t} = \frac{\partial}{\partial t}\left(\frac{2\lambda D}{d}\right) = 2\lambda D \times \frac{\partial}{\partial t}\left(d^{-1}\right)$$

$$= -\frac{2\lambda D}{d^2}\partial d\frac{\partial t}{}$$

$$= -\frac{2\lambda D}{d^2}\left[0.04\omega \u2061 \cos \u2061 \omega t\right] \times 10^{-3}$$

$$V_{\max \u2061} = \frac{-2\lambda D}{d^2}40 \times 10^{-5}$$

$$= \frac{2 \times 6 \times 10^{-7} \times 1 \times 4 \times 8 \times 10^{-7}}{(0.8 \times 10^{-3})^2}$$

$$= \frac{12 \times 32 \times 10^{-14}}{64 \times 10^{-8}} = 6\mu m / s$$

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